24.5.2016 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

Max. Marks: 100

April-May 2016

End Semester Exam

Duration:3 hrs

Class: SE

Name of the Course: Data Structures

Course Code: **UCEC303**

Semester: III

Branch: Computer

MUCBGS

Instructions:

(1) All Questions are Compulsory
(2) Make Suitable assumption whenever necessary and justify the same.
(3) Figures at the right indicate full marks.
(4) Specify the question numbers correctly.

Question No.		**Max. Marks**
Q1	a) Write a Program to implement singly linked list. b) Show the formation of Binary Search Tree for the following elements 45,39,56,12,34,78,32,10,89,54,67,81. Or b) Write a program to accept a graph and perform depth first traversal of the graph.	(10 + 10)
Q2	a) Write a program to create a sorted doubly linked list and perform following operations on it. i) Insert into the list ii) Delete from the list Or a) Write a program to implement Construction of Expression tree and its evaluation. b) Write a program to implement Output Restricted deque. Or b) Write a program to implement two stacks using array.	(10 +10)
Q3	a) Write the program to convert infix expression to prefix form. b) Write an algorithm for addition and multiplication of sparse matrices Or b) Explain Johnson's algorithm with an example.	(10 + 10)
Q4	a) Write the program to perform merge sort. Show the steps with the example	(10 + 10 + 10)

	Or a) Write a program to implement interpolation search. Explain with an example b) Show the hash table entries (size=11) for the given data using Double Hashing method 72,27,36,24,63,81,92,101 b) Sort the following elements in ascending order using Quick sort 77,49,25,12,9,33,56,81	
Q5	a) Identify the appropriate data structure to check whether a string is palindrome or not. Justify the same. Write the algorithm to for this operation Or b) Identify the appropriate data structure given the String containing arithmetic expression, to check whether opening and closing parenthesis are well formed. Justify the same. Write the algorithm for this operation	(10)